

## Problem Set 12

### Odd, Even, and Periodic Functions

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#### Problem 2

(a)

$$\begin{aligned} f(x) &= -f(-x), \quad g(x) = -g(-x) \\ h(x) &= f(x)g(x) \\ h(-x) &= f(-x)g(-x) = (-f(x))(-g(x)) = h(x) \end{aligned}$$

$h(x)$  is even

$$\begin{aligned} f(x) &= -f(-x), \quad h(x) = h(-x) \\ k(x) &= f(x)h(x) \\ k(-x) &= f(-x)h(-x) = -f(x)h(x) = -k(x) \end{aligned}$$

$k(x)$  is odd

(b)

$$\begin{aligned} f(x) &= f(-x) \\ f'(x) &= -f'(-x) \end{aligned}$$

$f'(x)$  is odd

$$\begin{aligned} g(x) &= -g(-x) \\ g'(x) &= g'(-x) \end{aligned}$$

$g'(x)$  is even

(c)

$$\begin{aligned} f(x) &= -f(-x) \\ g(x) &= \int_a^x f(t) \, dt \\ g'(x) &= f(x) \\ h(x) &= g(x) - g(-x) \\ h'(x) &= g'(x) + g'(-x) = f(x) + f(-x) = 0 \\ h(0) &= g(0) - g(0) = 0 \\ g(x) - g(-x) &= 0 \implies \boxed{g(x) = g(-x)} \end{aligned}$$

(d)

$$\begin{aligned}
a = 0 &\implies F(x) = \int_0^x f(t) dt \\
f(t) &= f(-t) \\
F(-x) &= \int_0^{-x} f(t) dt = - \int_0^x f(-x) dx = - \int_0^x f(x) dx \\
&\boxed{F(x) = -F(-x)} \\
a \neq 0, f(x) &= 1 \\
F(x) &= \int_1^x f(t) dt = x - 1 \\
F(-x) &= -x - 1 \\
&\boxed{F(x) \neq -F(-x)}
\end{aligned}$$

(e) No, only one where the constant  $C = 0$  is odd.**Problem 3**

(d)

$$\begin{aligned}
&\int_{-2}^2 \frac{\arcsin(\sin x) + \arccos(\cos x)}{(1+x^2)(1+\lfloor \frac{x^2}{5} \rfloor)} dx \\
&= \int_{-2}^2 \frac{\arcsin(\sin x)}{(1+x^2)(1+\lfloor \frac{x^2}{5} \rfloor)} dx + \int_{-2}^2 \frac{\arccos(\cos x)}{(1+x^2)(1+\lfloor \frac{x^2}{5} \rfloor)} dx \\
&= 0 + 2 \int_0^2 \frac{\arccos(\cos x)}{(1+x^2)(1+\lfloor \frac{x^2}{5} \rfloor)} dx \\
&= 2 \int_0^2 \frac{x}{1+x^2} dx \\
&= \int_1^5 \frac{1}{u} du \\
&= \ln 5 - \ln 1 \\
&= \boxed{\ln 5}
\end{aligned}$$

**Problem 4**

(a)

$$\begin{aligned}
E(x) &= \frac{f(x) + f(-x)}{2} \\
E(-x) &= \frac{f(-x) + f(-(-x))}{2} = \frac{f(x) + f(-x)}{2} \\
&\boxed{E(x) = E(-x)}
\end{aligned}$$

$$\begin{aligned}
 O(x) &= \frac{f(x) - f(-x)}{2} \\
 O(-x) &= \frac{f(-x) - f(-(-x))}{2} = \frac{f(-x) - f(x)}{2} \\
 &\quad \boxed{O(x) = -O(-x)} \\
 f(x) &= \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2} = E(x) + O(x)
 \end{aligned}$$

(b)

$$\begin{aligned}
 \cosh x &= \frac{e^x + e^{-x}}{2} \\
 \sinh x &= \frac{e^x - e^{-x}}{2}
 \end{aligned}$$

(c)

$$\int_{-a}^a f(x) dx = \int_0^a f(x) dx + \int_{-a}^0 f(x) dx = \int_0^a [f(x) + f(-x)] dx$$

(d)

$$\begin{aligned}
 \frac{1}{1+e^x} + \frac{1}{1+e^{-x}} &= \frac{2+e^x+e^{-x}}{2+e^x+e^{-x}} = 1 \\
 \int_{-1}^1 \frac{dx}{1+e^x} &= \int_0^1 \frac{1}{1+e^x} + \frac{1}{1+e^{-x}} dx = \int_0^1 1 dx = \boxed{1}
 \end{aligned}$$

(e)

$$\begin{aligned}
 \frac{x^2}{1+e^{x^3}} + \frac{x^2}{1+e^{-x^3}} &= x^2 \\
 \int_{-2}^2 \frac{x^2}{1+e^{x^3}} dx &= \int_0^2 \frac{x^2}{1+e^{x^3}} + \frac{x^2}{1+e^{-x^3}} dx = \int_0^2 x^2 dx = \boxed{\frac{8}{3}}
 \end{aligned}$$

## Problem 5

(a)

$$\begin{aligned}
 \int_a^{a+T} f(x) dx &= \int_a^T f(x) dx + \int_T^{a+T} f(x) dx \\
 &= \int_a^T f(x) dx + \int_0^a f(x) dx = \int_0^T f(x) dx
 \end{aligned}$$

(b)

$$\int_0^{nT} f(x) dx = \sum_{k=0}^{n-1} \int_{kT}^{(k+1)T} f(x) dx = n \int_0^T f(x) dx$$

(c)

$$\begin{aligned}
 T_{|\sin x|} &= \pi \\
 \int_0^{\frac{16\pi}{3}} |\sin x| dx &= 5 \int_0^{\pi} |\sin x| dx + \int_0^{\frac{\pi}{3}} |\sin x| dx \\
 &= 5 \cdot 2 + 1 - \frac{1}{2} = \boxed{\frac{21}{2}}
 \end{aligned}$$

(d)

$$\begin{aligned}
 F(x+T) - F(x) &= \int_0^{x+T} f(t) dt - \int_0^x f(t) dt = \int_x^{x+T} f(t) dt = \int_0^T f(t) dt \\
 F(x+T) = F(x) &\implies F(x+T) - F(x) = 0 \\
 &\iff \int_0^T f(t) dt = 0
 \end{aligned}$$

(e) As  $\int_0^T f(t) dt \neq 0$ , the shape of  $F(x)$  is not periodic.**Problem 7**

(a)

$$\begin{aligned}
 t \in \{0, 2\pi, 4\pi\} &\implies y = 1 \\
 t \in (0, \frac{\pi}{2}] \cup [\frac{3\pi}{2}, 2\pi) \cup (2\pi, \frac{5\pi}{2}] \cup [\frac{7\pi}{2}, 4\pi) &\implies y = 0 \\
 t \in (\frac{\pi}{2}, \frac{3\pi}{2}) \cup (\frac{5\pi}{2}, \frac{7\pi}{2}) &\implies y = -1 \\
 T &= 2\pi
 \end{aligned}$$

(b)

$$\int_0^{2\pi} \lfloor \cos x \rfloor dx = -\pi$$

The isolated point does not matter, as its width is too insignificant compared to the rest of the function.

(c)

$$\int_0^x \lfloor \cos t \rfloor dt = \int_0^{2\pi n} \lfloor \cos t \rfloor dt = n \cdot -\pi = \boxed{-n\pi}$$

(d)  $G$  is constant on the intervals where  $y = 0$  and decreasing on the intervals where  $y = -1$ .

(e)

$$\begin{aligned}
 T &= 1 \\
 \int_0^{7.5} s(t) dt &= 7 \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{4} = \boxed{\frac{29}{8}}
 \end{aligned}$$

**Problem 8**

(a)

$$\begin{aligned}\frac{g(x)}{1+b^x} + \frac{g(x)}{1+b^{-x}} &= g(x) \cdot \frac{2+b^x+b^{-x}}{2+b^x+b^{-x}} = g(x) \\ \int_{-a}^a \frac{g(x)}{1+b^x} dx &= \int_0^a \frac{g(x)}{1+b^x} + \frac{g(x)}{1+b^{-x}} dx = \int_0^a g(x) dx\end{aligned}$$

(b)

$$\begin{aligned}I_n &= \int_0^\pi \frac{\sin nx}{\sin x} dx \\ I_{n+2} - I_n &= \int_0^\pi \frac{\sin((n+2)x) - \sin(nx)}{\sin x} dx \\ &= \int_0^\pi 2 \cos((n+1)x) dx \\ &= \frac{2}{n+1} \sin((n+1)x) \Big|_0^\pi = 0 \\ &\quad \boxed{I_{n+2} = I_n} \\ I_1 &= \int_0^\pi 1 dx = \pi \\ I_2 &= \int_0^\pi 2 \cos x dx = 0\end{aligned}$$

(c)

$$\begin{aligned}\int_{-\pi}^\pi \frac{\sin(2027x)}{(1+2^x)\sin x} dx &= \int_0^\pi \frac{\sin 2027x}{\sin x} = \boxed{\pi} \\ \int_{-\pi}^\pi \frac{\sin(2026x)}{(1+2^x)\sin x} dx &= \int_0^\pi \frac{\sin 2026x}{\sin x} = \boxed{0}\end{aligned}$$

(d) The bases cancel out upon combining two intervals together for an even function.